THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The HCF of 24, 36 and 60 is: 1

(a) 24 **(b)** 360 **(c)** 12 **(d)** 6

2. If α, β are zeroes of $x^2 - 7x + 10$, then $\alpha^2 + \beta^2$ is: 1

(a) 69 **(b)** 49 **(c)** 29 **(d)** 39

3. The value of x in $7x + 2y = -6$, $3x + 5y = 14$ is: 1

(a) -8 **(b)** 2 **(c)** 4 **(d)** -2

4. The sum of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) 2 **(b)** -2 **(c)** $-\frac{9}{2}$ **(d)** $\frac{9}{2}$

5. The 12th term of the AP whose 1st term is 7 and 5th term is 19 is: 1

(a) 40 **(b)** 37 **(c)** 36 **(d)** 43

6. If $DE \parallel BC$ in $\triangle ABC$, $AB = 5$ cm, $AD = 3$ cm, $AC = 10$ cm, then EC (cm) is: 1

(a) 6 **(b)** 10 **(c)** 4 **(d)** 2

7. The perimeter of the triangle with vertices $(0, 0)$, $(6, 0)$, $(0, 8)$ is: 1

(a) 34 **(b)** 24 **(c)** 14 **(d)** 28

8. The value of $\sin^2 30^\circ + \sin^2 45^\circ$ is: 1

(a) $\frac{7}{4}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{3}{2}$ **(d)** $\frac{1}{4}$

9. A boy whose eyes are 2 m above the ground stands 18 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) $20\sqrt{3}$ m **(b)** 21 m **(c)** 20 m **(d)** 18 m

10. The length of the tangent from an external point P to a circle is 16 cm and P is 34 cm from the centre. The radius of the circle is: 1

(a) 18 cm **(b)** 50 cm **(c)** 30 cm **(d)** 17 cm

11. A square is inscribed in a circle of radius 7 cm. The area of the circle outside the square is: 1

(a) 154 cm² **(b)** 98 cm² **(c)** 56 cm² **(d)** 105 cm²

12. A tent is a cylinder of radius 3.5 m and height 4 m surmounted by a cone of the same radius and height 12 m. The area of canvas (in m²) needed to make the tent is: 1

(a) 137.5 m² **(b)** 220 m² **(c)** 225.5 m² **(d)** 88 m²

13. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 6, 9, 7, 5, 16, 7 respectively, The mean of the data is: 1

(a) 270 **(b)** 33.4 **(c)** 35 **(d)** 32.4

14. A red die and a blue die are rolled together. The probability that the sum is a multiple of 4 is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{5}{18}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{3}{4}$

15. The LCM of 6, 8 and 12 is: 1

(a) 24 **(b)** 48 **(c)** 12 **(d)** 30

16. If α, β are zeroes of $x^2 - 5x + 6$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is: 1

(a) $\frac{5}{6}$ **(b)** 30 **(c)** $-\frac{5}{6}$ **(d)** $\frac{6}{5}$

17. The value of y in $11x + 4y = 53$, $5x + 3y = 30$ is: 1

(a) -5 **(b)** 8 **(c)** 3 **(d)** 5

18. The product of the roots of $2x^2 - 9x + 4 = 0$ is: 1

(a) $\frac{9}{2}$ **(b)** $-\frac{9}{2}$ **(c)** -2 **(d)** 2

19. The number of terms in the AP $5, 9, \ldots, 97$ is: 1

(a) 24 **(b)** 23 **(c)** 25 **(d)** 26

20. The areas of two similar triangles are 36 cm² and 100 cm². The ratio of their perimeters is 3: 1

(a) 15 **(b)** 25 **(c)** 5 **(d)** 8

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the largest number which divides 245 and 1029 leaving remainders 5 and 5 respectively. 2

22. If α, β are zeroes of $x^2 - 8x + 12$, find $\alpha^2 + \beta^2$. 2

23. Solve by the cross-multiplication method: $7x + 2y = -6$ and $3x + 5y = 14$. 2

24. Solve for x: $2x^2 - 9x + 4 = 0$ by the quadratic formula. 2

25. Which term of the AP $7, 13, 19, 25, \ldots$ is 127? 2

26. In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 5$ cm, $AD = 3$ cm and $AC = 10$ cm, find AE and EC. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. The HCF of two numbers is 9 and their LCM is 360. If one number is 72, find the other. 3

28. If α, β are zeroes of $x^2 - 6x + 8$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ and $\alpha^2 + \beta^2$. 3

29. Solve by elimination: $7x + 2y = -6$ and $3x + 5y = 14$. 3

30. Solve for x: $\frac{1}{x} + \frac{1}{x+1} = \frac{5}{6}$. 3

31. In an AP, the 3rd term is 19 and the 8th term is 54. Find the AP and its 15th term. 3

OR

In $\triangle ABC$, D and E lie on AB and AC with $DE \parallel BC$. If $AB = 9$ cm, $AD = 4$ cm and $AC = 18$ cm, find AE and EC. 3

MASTERMIND

32. Show that the points $(0,\ 0)$, $(4,\ 0)$ and $(2,\ 3)$ are the vertices of an isosceles triangle. 3

OR

If $\sin(3x^\circ) = \frac{1}{2}$, find the value of x (an acute angle). 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Two tankers contain 850 litres and 680 litres of fuel. Find the maximum capacity of a container that measures both exactly. How many measures does each tanker need? 5

34. Find all the zeroes of $x^3 - 6x^2 + 11x - 6$ if two of its zeroes are 1 and 2. 5

35. A father is 10 times as old as his son. After 6 years the father will be 4 times as old as the son will be. Find their present ages. 5

36. A train travels 200 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 2 hours less for the journey. Find the original speed. 5

OR

How many terms of the AP $54, 50, 46, 42, \ldots$ must be taken so that the sum is 360? 5

37. Two poles of heights 14 m and 2 m stand on level ground 9 m apart. Find the distance between their tops. 5

OR

In what ratio does the x-axis divide the segment joining $(-3,\ -5)$ and $(6,\ 4)$? Find the point of division. 5

38. Prove that: $(\sec A - \tan A)^2 = \dfrac{1 - \sin A}{1 + \sin A}$. 5

OR

Two vertical poles of equal height stand on opposite sides of a road 60 m wide. From a point on the road between them, the angles of elevation of the tops of the poles are 60° and 30°. Find the common height of the poles and the distances of the point from the poles (nearer pole first). 5

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND